

The Outcomes of the 2-Decade Monthly Monitoring of Fly Ash-Composition in a Lignite-Fired Power Station

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Received: 27 February 2010 / Accepted: 3 September 2010 / Published online: 26 September 2010
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Abstract In the current study, the evolution of the composition of fly ash (FA), since the beginning of operation of the lignite-fired power station of Amynteon-Filotas [Steam Electric Station (SES) Amynteon-Filotas] in Greece, is presented. The respective evolution of the feedcoal-composition is also showed. This power station began to operate in 1987. Before that, it was predicted that lignite fly ash would turn from strongly calcareous to barely pozzolanic. Moreover, the variation of sulphur (%) (both total and volatile) in feedcoal had revealed that fly ash would not often meet the specifications to be used in the production process of concrete, due to the excessive percentage presence of SO_3 . These predictions were eventually confirmed in the 23 years of operation of the power station. During these years SO_3 (%) kept alternating, rendering—in some cases—fly ash suitable for use in cement industry

applications. The periodical changes of the SO_3 concentration in fly ash were attributed to the free (volatile) sulphur of feedcoal as well as to the excess of FeS_2 , which had also a serious effect on the daily power production process (severe erosion of the metallic surfaces of boiler). A number of solutions for tackling this phenomenon are suggested in this study: discard of specific FA grain fractions, intense monitoring of the material and its effective mixture-homogenization and selective mining of lignite.

Keywords Fly ash · Lignite · Lignite-xylite co-firing · Hellenic thermal power stations · Specific consumption

Introduction

The lignite-fired power station of Amynteon-Filotas in Amynteon, Greece ($2 \times 300\text{MW}$) operates since 1987. For the power production process, lignite from the open-pit mine of Amynteon is fired. The mining- and power station-facilities are located in Western Macedonia, Northwestern Greece in a huge-aquifer-region (Lakes: Vegorititis, Chimaditis, Zazaris, Petron). The lignite deposit of the Ptolemais-Amynteon basin extends up to the Polifytos Lake to the south and further than the Hellenic borders to the north, within FYROM. Lignite of Amynteon was formed later than lignite of Ptolemais (at Pleistocene and Pliocene, respectively). Before the formation of lignite, water flow took place within the Amynteon basin, followed by the subsequent accumulation of sediments. Those included sand, which was concentrated into dunes within the confines of the deposit, as well as in separate layers of the lignite field (such as fine-grained sandy marl). The accumulation of sediments led to the variations of fuel-composition, which were afterwards reported [1].

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The chemical and mineralogical composition of the inorganic fraction which is included in the organic part of lignite (inherent ash) is considerably different from that of the inorganic fraction that comes along with the fuel after the mining process (extraneous ash) [2]. Most of the lignite fly ash (FA) particles consist of non-volatile compounds. Actually, the volatiles can either condensate in them, or form finer FA particles that are not trapped by the electrostatic precipitators [3]. The chalcophile (Goldschmidt) trace elements participate in that process by being enriched in the finer particles. Rather, the lithophile (Goldschmidt) trace elements are intensely presented in the coarse-grained fraction. Other components, with a diameter of approximately 0.1 μm , are incorporated in the so-called “pockets” of the fuel and they are then able to coalesce and to consequently form coarser particles [4–6].

The mineralogical fraction that is co-extracted with lignite varies as a function of the prevalent conditions throughout mining. The feedcoal of a power station normally contains various components that may participate in physical–chemical procedures. Such procedures also depend upon the specific conditions occurring during the power production process, such as standard lignite firing as in the present case, co-firing with SRF [7, 8] and CCS technologies in PF or CFB boilers [9–12]. Therefore, the composition of fly ash produced in a power station gradually changes. In some cases, that change is so profound that fly ashes from the same station turn, over the years, from Class C (ASTM C618) to Class F and vice versa. The variation of parameters that influence the process of lignite combustion often results in phenomena with serious consequences on the production process such as erosion of the metallic surfaces of boiler [13–15].

Methods

Composition of fuel and fly ash are monitored within the power production process of Amynteon-Filotas thermal power station by conducting the respective analyses according to the protocols showed in Table 1.

Table 1 ASTM protocols of the analyses conducted throughout the power production process

Analysis	Protocol
Air-dried moisture of lignite	ASTM D3173-73
Ash content of lignite	ASTM D3174-89
Volatile matter content of lignite	ASTM D3175-89
Calorific value of lignite	ASTM D2015-85 & D3286-85
Fly ash chemical composition	ASTM C311
Type of fly ash	ASTM C618

Proximate and Ultimate Analysis of Lignite

Proximate analysis is a fundamental assay in all coal quality assessments. Air-dried moisture, ash and volatile matter contents are determined directly. Air-dried moisture is obtained by heating the lignite sample at 105–110°C in (oxygen-free) nitrogen atmosphere (ASTM D3173-73). The loss in weight, expressed as a percentage, is taken as the air-dried moisture. The ash content is the weight of inorganic residue (expressed as a percentage) left after incineration at 750–815°C to constant weight in a ventilated furnace (ASTM D3174-89). The ash content is important because it reduces the heating value of a coal. It represents a “dead weight” during transportation and it gives an indication of the “dirtiness” of a coal.

The volatile matter-content of coal is determined by the “Automatic Multiple Sample Thermogravimetric Analyzer (TGA-2000) according ASTM (heating the sample within a platinum crucible in the absence of air at 930°C, ASTM D3175-89). The percentage weight loss, minus a correction for the air-dried moisture content, gives the volatile matter content. The volatiles liberated are predominantly carbon monoxide, carbon dioxide, methane, hydrogen, water and tar vapors derived from the organic coal structure. In addition, there is a small, but significant, volatile contribution from associated minerals.

The calorific value is determined at a constant volume by burning a known weight of the sample in a bomb calorimeter under conditions of excess oxygen, and measuring the amount of heat evolved (ASTM D2015-85 & D3286-85). Gross specific energy value may be misleading when applied to actual coal combustion, since in an operating boiler the combustion is unconfined and the water vapor produced will not be condensed. This results in a loss due to the latent heat. It has consequently become a standard practice to express the heating value as the net specific energy by correcting for the latent heat of combustion products. The net value is calculated from the experimentally determined gross specific energy at constant volume.

The ultimate analysis is a collective term for the determination of the main elements within a coal sample, i.e. carbon, hydrogen, nitrogen, sulphur and oxygen. Ultimate analysis is carried out using LECO C-H-N 1000 and LECO CS elemental analyzers.

Chemical Composition of Lignite Fly Ash

The (%) concentration of the major elements of lignite fly ash is determined by means of X-Ray Fluorescence (XRF), using a Spectro X-Lab 2000 Energy Dispersive XRF-spectrometer applying the samples in a pressed powder form. According to ASTM Standards (ASTM C618), fly

ash is classified based on the extent of its major elements (Si, Al and Fe) as Class C, F and N.

Results and Discussion

Fuel

Table 2 shows the annual average composition of feedcoal from 1987 to 2008. Figure 1 illustrates the variation of net calorific value, specific consumption and ash load throughout the first two decades of operation of the power station.

Since 1987, the quality of Amynteon lignite gradually worsens in terms of its increased ash content and decreased calorific value. Therefore, lignite from the region of Ptol-emaïs is often used in the fuel-mix since 1990. Since 1992, xylite from the region of Florina is systematically co-fired with lignite (4–10% to the fuel-mix), thus resulting in the increased calorific value of feedcoal.

In the period 2000–2007, the annual average net calorific value (kcal/kg) gradually increased from 1150 to 1350 kcal/kg (average annual increase ~ 25 kcal/kg). The average annual variation of net calorific value included a much bigger variation of its monthly values, ranging from

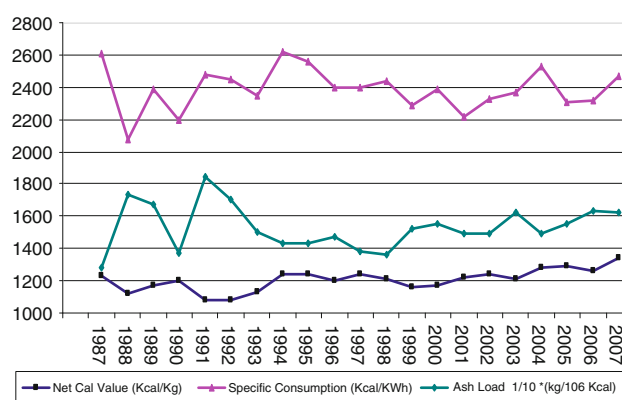


Fig. 1 Net calorific value, specific consumption and ash load throughout the 2 decades of operation of the Amynteon-Filotas power plant

50 to 150 kcal/kg/month. The respective variation in a daily basis was often even bigger. Within the same period, the average annual ash content (%) gradually increased from 19 to 21%, while the average percentage presence of sulphur (air-dried basis) varied, with the tendency to reach the point of 1%. The average percentage presence of sulphur was bigger in the first period of operation of the power plant (1.8%).

Table 2 Annual average composition of feedcoal

Year	Air-dried basis					As received basis	
	Volatile matter (%)	Ash (%)	C (%)	H (%)	S (%)	Total moisture (%)	Net cal value (kcal/kg)
1987	39.49	36.2	40.80	3.56	1.58	57.0	1251
1988	37.28	41.29	37.47	3.25	1.56	54.75	1122
1989	37.80	41.29	35.00	2.98	1.60	54.35	1169
1990	36.78	41.73	36.03	3.19	1.81	52.93	1193
1991	37.92	41.99	33.24	3.22	1.45	53.67	1074
1992	39.46	40.15	34.11	3.16	1.19	53.72	1093
1993	39.99	38.56	35.66	3.24	1.17	54.04	1134
1994	40.34	37.23	35.63	2.84	1.20	53.63	1215
1995	41.83	37.76	36.65	2.90	1.25	53.29	1234
1996	43.27	38.11	32.64	2.96	1.15	53.80	1197
1997	43.41	37.14	32.42	3.02	1.22	53.58	1244
1998	43.78	36.08	33.98	3.06	1.12	54.74	1207
1999	43.20	37.36	33.52	2.89	1.35	54.10	1183
2000	40.02	40.89	29.97	2.74	1.29	52.76	1146
2001	42.04	37.66	32.48	2.90	1.10	52.14	1227
2002	42.18	38.12	36.35	2.91	1.09	51.83	1251
2003	41.01	39.56	33.88	2.84	1.12	50.87	1220
2004	41.01	40.80	35.36	2.94	1.13	50.85	1176
2005	35.33	39.29	35.50	2.97	1.18	49.54	1298
2006	51.26	36.19	36.40	3.84	0.98	51.26	1275
2007	35.41	36.64	34.52	2.34	1.05	46.65	1398
2008	33.52	42.71	32.74	1.76	1.01	46.02	1302

Regarding maceral composition of lignites from Ptolemais/Amynteon basin, Papanicolaou et al. [16] report that they mainly contain: Eulminite B, Humotelinite, Attrinite, Densinite, Humodrinite (52 vol%), Corpohuminite, Levigelinite, Humocollinite (Total Huminite: 80 vol%), Funitite, Inertodetrinite, Sclerotinite (Total Inertinite 17 vol%). Although the (%) composition of the maceral groups is relatively the same in all Greek coal basins, the (%) composition of maceral subgroups, especially that of huminite, substantially varies. This variation affects the performance of boiler since the maximum heat-release can take place at various heights inside the boiler. Humotelinitic brown coals rich in resinite release their thermal energy at lower temperatures than humocollinitic coals, which have low resinite contents.

Fly Ash

Figures 2, 3, 4, 5, 6, and 7 show the differentiation of the concentration of specific components of fly ash during the first two decades of operation of SES Amynteon-Filotas. Table 3 shows the annual average composition of fly ash from 1987 to 2008.

For the production process of the power station, along with lignite from the region of Amynteon, lignite from the region of Ptolemais was also used in December '90, February '92, May '93, August '93, September '93 and June '95. Partially due to the co-firing of Ptolemais and Amynteon lignite, fly ash with a very high weight percentage of CaO was produced from 1988 to 1997. At the same time, the percentages of SiO₂ and SO₃ in fly ash were lower compared to the respective values during the second decade of the power station-operation. CaO in fly ash peaked in September '93 (36.02%); fly ash of September '93 also presented the second lowest monthly percentage of SiO₂ to date (31.30%). On the other hand, co-firing of lignite and xylite during the second decade resulted in more siliceous

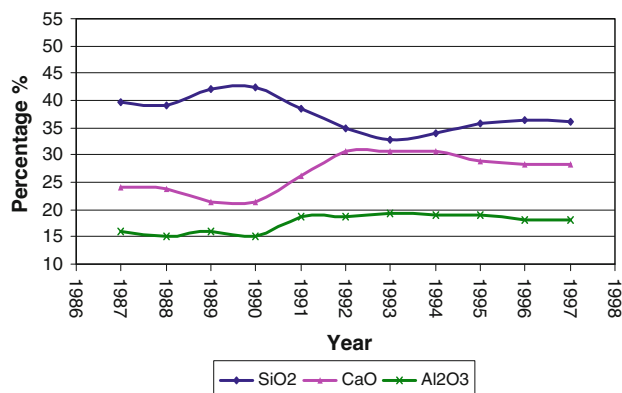


Fig. 2 Variation of SiO₂, CaO and Al₂O₃ (%) during the first decade of operation of the Amynteon-Filotas power plant

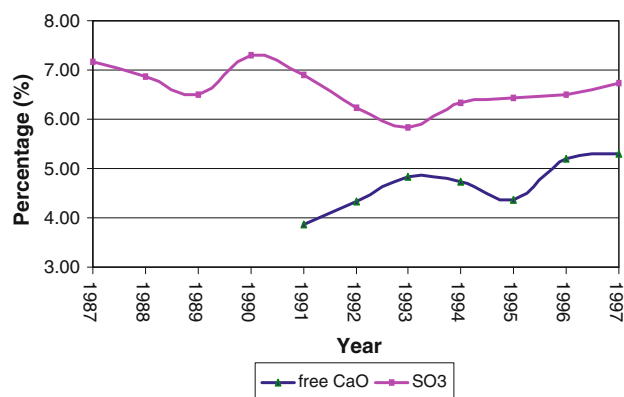


Fig. 3 Variation of free CaO and SO₃ (%) during the first decade of operation of the Amynteon-Filotas power plant

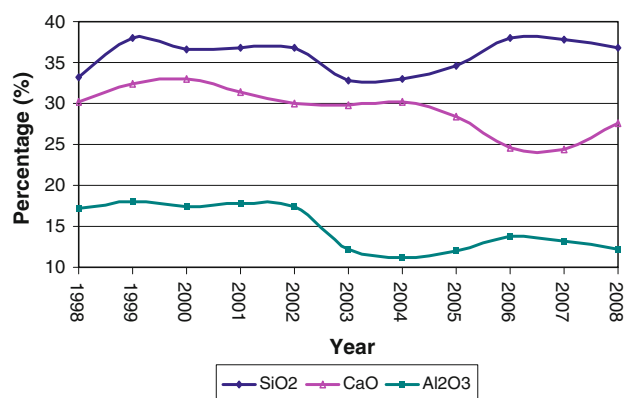


Fig. 4 Variation of SiO₂, CaO and Al₂O₃ (%) during the second decade of operation of the Amynteon-Filotas power plant

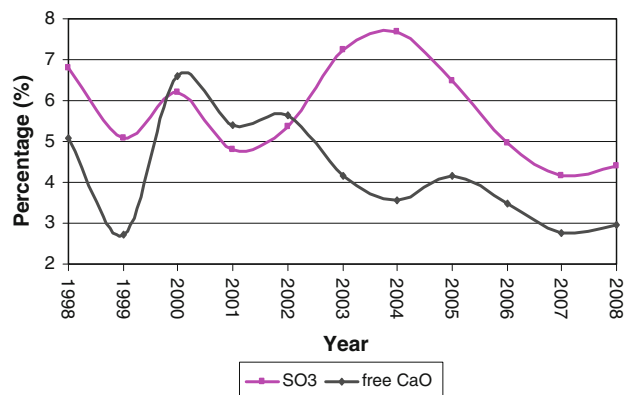


Fig. 5 Variation of free CaO and SO₃ (%) during the first decade of operation of the Amynteon-Filotas power plant

fly ash (SiO₂: 41.00%, February 1997) with low percentage of CaO (23.36%). The highest SiO₂ percentage (1989, aa: 42.16%) in fly ash was reported when lignite of Amynteon was exclusively burned. At the same period, the lowest mean annual percentage of CaO in fly ash was also reported (aa: 20.67%, year: 1990). Low CaO was followed

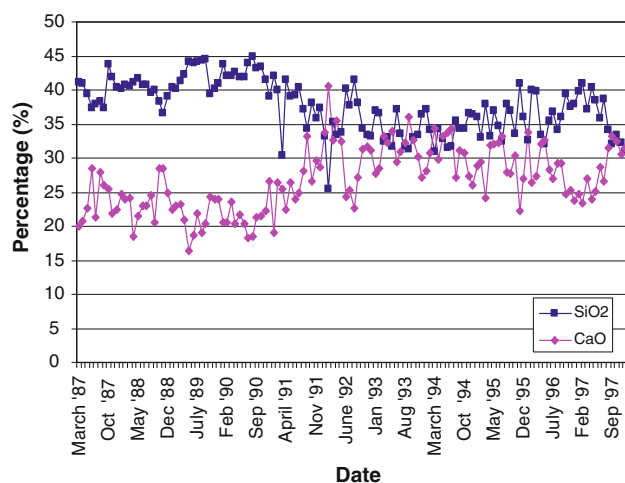


Fig. 6 Monthly variation of SiO_2 and CaO (%) during the first decade of operation of SES Amynteon-Filotas

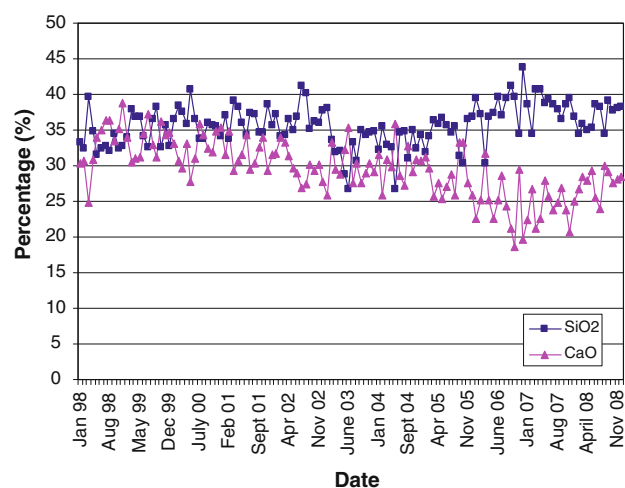


Fig. 7 Monthly variation of SiO_2 and CaO (%) during the second decade of operation of SES Amynteon-Filotas

by high SO_3 and low Al_2O_3 . Moreover, high sulphur percentages in fly ash came along with high Fe_2O_3 (%), an indicator of iron pyrite (FeS_2) being present in feedcoal during the first decade of operation of the power station.

The decline of Fe_2O_3 in fly ash from $\sim 8\%$ to $\sim 4\%$ after 1990 came along with the reduction of total sulfur in feedcoal. In fact, the abrasion caused by the impact of the hard components of ash (quartz and iron pyrite) on the iron surfaces of boiler, during lignite burning, had been resulting in the enrichment of metallic Fe in fly ash, due to the loss of material of the metallic boiler surfaces, until 1990. SO_3 (%) in fly ash was also slightly reduced; SO_3 was attributed not only to FeS_2 , but also to organic sulphur and sulphuric salts of feedcoal. Moreover, the 5-percent-increase of CaO in ash during the same period, which came along with a 2-percent-increase of CO_2 in feedcoal, was

Table 3 Average annual composition of fly ash throughout the 2 decades of operation of the Amynteon-Filotas power plant

Year	SiO_2	Fe_2O_3	Al_2O_3	TiO_2	CaO	MgO	SO_3	Na_2O	K_2O
1987	39.90	7.83	15.94	0.58	23.70	3.07	7.04	0.76	0.85
1988	39.88	8.00	15.22	0.76	23.87	3.42	6.88	0.69	1.03
1989	42.16	8.53	16.01	0.74	21.53	3.00	6.49	0.70	0.96
1990	41.29	8.45	15.67	0.80	20.67	2.87	6.98	0.62	0.98
1991	38.63	3.79	18.56	0.95	26.21	3.74	6.90	0.69	0.83
1992	34.99	3.43	18.67	0.98	30.71	3.65	6.21	0.68	0.85
1993	33.92	3.61	19.42	0.97	31.17	4.03	5.84	0.59	0.94
1994	34.12	3.59	19.00	0.98	30.56	3.87	6.34	0.66	0.85
1995	35.81	3.86	19.10	0.97	28.93	3.12	6.42	0.87	0.92
1996	36.27	3.79	18.14	0.92	25.67	3.67	6.49	1.30	1.10
1997	36.19	3.62	18.03	0.78	28.24	3.48	6.73	1.58	1.34
1998	33.54	3.64	16.97	0.78	33.32	2.34	6.47	1.60	1.35
1999	38.06	3.32	17.98	0.86	32.37	1.00	5.07	0.66	0.68
2000	36.63	3.61	17.36	0.73	33.05	2.13	6.20	0.14	0.15
2001	36.38	3.63	17.68	0.82	31.62	3.63	3.53	0.82	0.75
2002	38.57	4.00	17.57	0.70	27.51	3.13	5.62	1.57	1.33
2003	31.79	7.09	11.34	0.75	30.11	3.80	6.92	2.24	2.05
2004	33.00	7.30	11.50	0.71	29.65	4.03	7.71	0.79	1.28
2005	31.31	7.42	11.48	0.70	30.04	3.83	7.47	0.87	1.19
2006	30.28	7.67	10.10	0.67	31.73	4.04	7.36	3.29	1.00
2007	34.47	7.70	11.86	0.74	26.68	3.75	6.77	3.25	1.28
2008	34.98	7.74	11.64	0.89	27.95	3.57	4.02	2.93	2.45

attributed to the intense presence of limestone (CaCO_3) in feedcoal; CaCO_3 is an easy-to-crumble mineral that reduces the abrasive action of ash particles colliding with the metallic surfaces of boiler. CaCO_3 is normally co-extracted with lignite and its thermal disruption in the combustion chamber leads to high free lime percentages in the produced fly ash, which subsequently results in its strong flue gases-desulphurization capacity. From 2001 to 2002, CaO and SO_3 consisted approximately 30% and 5% of the chemical composition of fly ash respectively (5.5% free CaO at the same period). Then, until 2003, the concentrated presence of SO_2 in flue gases (due to the increase of volatile sulphur in feedcoal) resulted in the reduction of calcium components in fly ash. Indeed, free CaO of fly ash decreased from 7% to 4% in 2003. Such components could have improved desulphurization during the daily power production process. Until 2004 the increased sulphur of feedcoal was leading to higher SO_3 (%) in fly ash (bigger than 7.5%) and subsequently to lower percentages of free CaO in fly ash (approximately 3.5%). From 2005 to date, lower SO_2 in flue gases has been resulting in lower SO_3 (%) and subsequently in the higher free lime (%) of fly ash. Indeed, SO_3 (%) declined after 1998. However, it has been periodically varying (approximately every 2 years), a fact which was attributed to a load of FeS_2 ; FeS_2 is co-extracted

with lignite and causes serious problems towards the daily power production process. The variation of SiO_2 (%) and Al_2O_3 (%) from 2000 to date is caused by the aluminosilicate components, while the variation CaO (%) is mainly attributed to the presence of limestone. CaCO_3 of feedcoal was much lower in the first decade of the power plant operation than in the second. Indeed, in the first 5 years of the power plant operation, free CaO (%) in fly ash was practically zero.

The high SO_3 percentages in fly ash until 1998 obstructed the utilization of the material in concrete-manufacturing applications. Since then, SO_3 (%) was remarkably lower, with the exception of the period 2003–2005.

The considerable monthly variation in the chemical composition of fly ash is not reflected when it is given in an annual basis. In fact, there is an even bigger variation in the composition of fly ash when this is considered in a daily basis (when fuel consumption reaches the point of approximately 22×10^3 tones/day). For example, the average CaO (%) in fly ash between March '06 and August '07 was approximately 24%, while at the same period the minimum average monthly CaO (%) was approximately 18% and the maximum, 29%. In the same period, free CaO (%) in fly ash was determined from 1 to 3% and SO_3 (%) in fly ash between 4 and 7%.

Chemical composition and physical properties significantly vary as a function of the particle size distribution of Amynteon lignite fly ash. According to Itskos et al. [17], a fine-grained fraction ($>50 \mu\text{m}$) of considerable amount presents properties that obstruct the sustainable exploitation of calcareous Amynteon lignite fly ash in cement industry applications. However, the same grain fraction, because of its hydraulic properties, can be utilized in other short of applications based on different criteria, such as road constructions. The coarse-grained fraction ($>150 \mu\text{m}$), which reflects a low proportion to the total fly ash output, presents unwanted characteristics as well. On the other hand, the intermediate grain fraction presents the strongly desirable properties when fly ash is utilized as pozzolanic additive. Moreover, the mechanism of the formation of the intermediate grain fraction strongly prevents the factors that cause the variation of the quality of fly ash.

Regarding some other types of Hellenic fly ashes, Tsimas and Moutsatsou-Tsima [12] report that: (a) in Megalopolis fly ash (barely Class C), SiO_2 ranges from 47 to 52%, CaO from 5 to 15% and Al_2O_3 from 12 to 22%. (b) in Kardias fly ash (Class C), SiO_2 ranges from 26 to 34%, CaO from 33 to 41% and Al_2O_3 from 12 to 17%. (c) in Ptolemais fly ash (Class C) SiO_2 ranges from 28 to 41%, CaO from 23 to 39% and Al_2O_3 from 13 to 19% and (d) in Agios Dimitrios fly ash (Class C) SiO_2 ranges from 33 to 42%, CaO from 29 to 38% and Al_2O_3 from 9 to 12%. The fact

that Hellenic ashes are mostly classified as Class C is mainly attributed to the presence of compounds such as $3\text{CaO} \cdot \text{Al}_2\text{O}_3$, $\text{CaO} \cdot \text{SiO}_2$, $4\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{SiO}_2$ and sulphur-calcium aluminates [18–22]. In the highly-calcareous fly ash from the region of Ptolemais, the index $K = \text{CaO} + \text{MgO} + \text{alkalies}/\text{SiO}_2 + \text{Al}_2\text{O}_3$ fluctuates between 0.2 and 1.0, a fact revealing that they have not only pozzolanic, but also hydraulic behavior. In all cases the sum of $\text{Al}_2\text{O}_3 + \text{SiO}_2 + \text{Fe}_2\text{O}_3$ is greater than 50% but less than 70%, as it happens for Class F fly ashes. In addition, in all cases SiO_2 is greater than free CaO ; the greater this difference is, the bigger is the tendency for the pozzolanic reaction to happen between reactive silica and the $\text{Ca}(\text{OH})_2$ produced during the hydration of the minerals phases of clinker. It is mentioned that only $\sim 15\%$ of the $\sim 12 \times 10^6$ tones of lignite fly ash are utilized in Greece; the most significant uses of ashes in Greece refer to cement industry applications, replacing cement clinker and aiming to the production of several CEM II types according to EN 197-1 standard [23]. Tsimas and Moutsatsou-Tsima [18] summarize the following reasons for this phenomenon: (1) the inhomogeneity in the chemical and mineralogical composition of fly ash (2) the necessity for supplementary grinding of ashes for better reveal of their pozzolanic and hydraulic properties (3) the elevated proportion of their free CaO as its hydration causes severe problems as well as temperature increase and (4) the periodically elevated proportions of SO_3 . Though this study, it has become clear that the aforementioned problems also apply to lignite fly ash from the Amynteon-Filotas thermal power plant.

Conclusions

The considerable deviation in the chemical composition of fly ash makes its intense monitoring necessary both in terms of its possible utilization and the confrontation of phenomena that obstruct the power production process, i.e. the erosion of the metallic surfaces of boiler. The systematic monitoring of fly ash-composition gives the opportunity of mixing different daily fractions of the material for its effective homogenization.

The constantly high SO_3 (%) in Amynteon fly ash obstructs its utilization as an additive material in concrete manufacturing. The following actions are suggested in order for this phenomenon to be mitigated:

- The rejection of the specific FA grain fraction that mainly increases the total SO_3 (%) of lignite fly ash. The implementation of cyclones is needed for the separation of fly ash into its fractions. Through this process, free CaO in fly ash can be also effectively controlled.

- The selective utilization of lignite and the rejection of the mineral fraction with a load of FeS_2 . However big the cost of such process may be, the mitigation of erosion- and abrasion-related problems of boiler is required, in order for the effectiveness of the every-day power production process to be enhanced.

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